

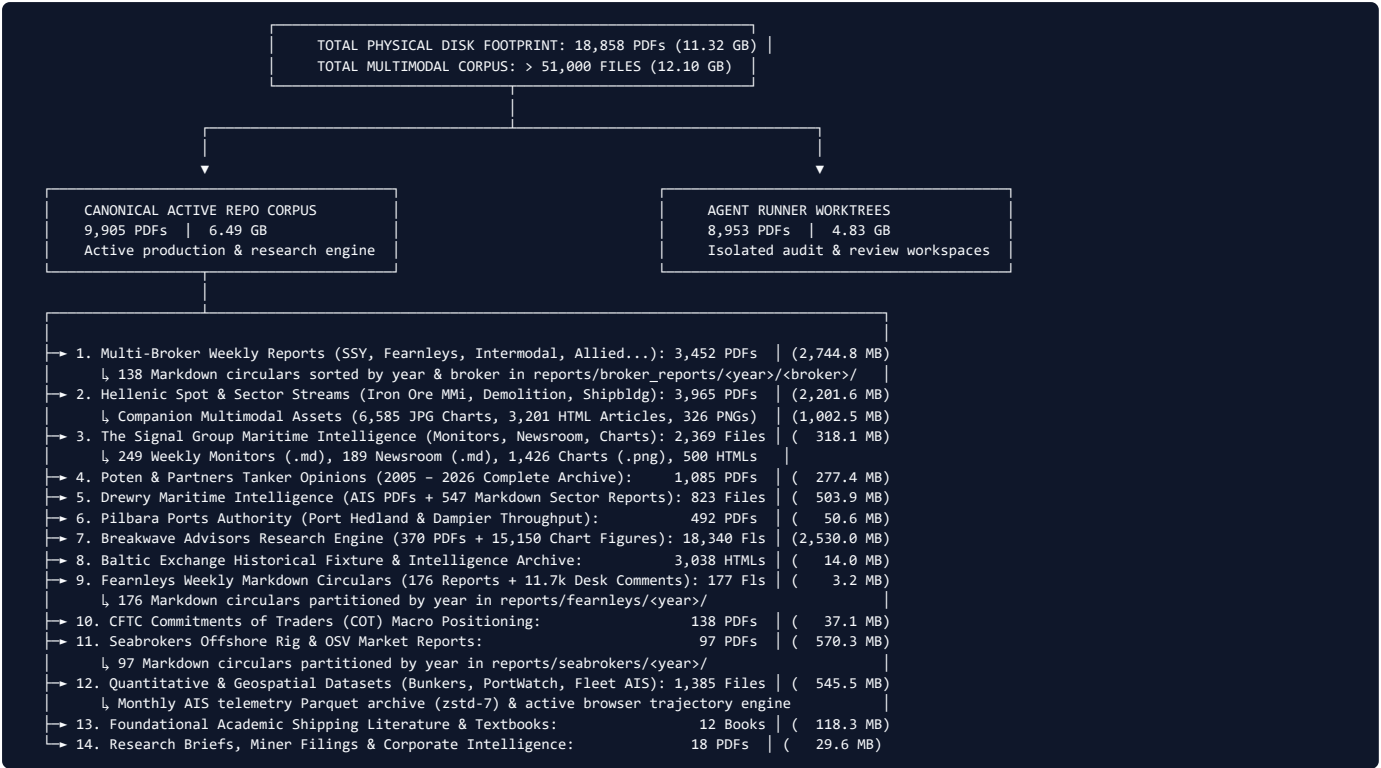
# Comprehensive Maritime Document & Multimodal Intelligence Corpus Audit

Repository: [Shipping](#) | Canonical Corpus Scope: Global Maritime Intelligence Architecture  
Audit Timestamp: September 20, 2026 | Target Artifact: Forensic Document Mapping, Multimodal Assets, AIS Trajectory Engine & Canonical Repository Architecture

## Executive Summary & High-Level State of the Corpus

The maritime analytics architecture hosts a multi-year, institutional-grade research corpus spanning **freight rates, vessel supply/demand, shipyard contracting, demolition/recycling, port throughput, charter fixtures, commodity trade flows, macro shipping intelligence, and quantitative time-series**.

While the primary document scan records **18,858 native PDFs (11.32 GB)** across canonical directories and agent runner worktrees, the actual intelligence architecture is **multimodal by design**. Over **51,000 files and 12.1 GB of structured data** are systematically harvested, processed into clean Markdown ( `.md` ), indexed into semantic knowledge trees ( `knowledge/trees/` ), visualized through high-resolution chart archives ( `.png` , `.jpg` ), or unpacked into tabular time-series ( `.csv` , `.json` , `.parquet` ).



## Complete Multimodal Asset Inventory (Comprehensive Repo Scan)

| Asset Class / File Type         | Extension      | Count  | Total Size             | Primary Role in System                                                              |
|---------------------------------|----------------|--------|------------------------|-------------------------------------------------------------------------------------|
| Native Document PDF             | .pdf           | 18,858 | 11,593.9 MB (11.32 GB) | Original scanned broker reports, bank research, textbooks, SEC filings              |
| Markdown Reports & Articles     | .md            | 38,280 | 754.2 MB               | Normalized text representations, opinion articles, knowledge docs, Signal monitors  |
| Visual Charts & Infographics    | .jpg / .jpeg   | 34,843 | 6,022.2 MB             | Embedded broker charts, Hellenic market figures, fleet layouts, port maps           |
| Diagrams & Vector Art           | .png / .avif   | 34,803 | 4,895.7 MB             | High-res freight charts, Signal Ocean curves, technical diagrams, route schematics  |
| Structured Metadata JSON        | .json          | 32,242 | 1,169.7 MB             | Parsed tables, document metadata, route matrices, desk summaries, commodity trees   |
| Original HTML Provenance Dumps  | .html          | 29,716 | 567.7 MB               | Raw scraped web articles from Signal Ocean, Hellenic, Baltic Exchange, Breakwave    |
| Master Tabular CSVs             | .csv           | 1,008  | 887.1 MB               | Time charter indices, port calls, iron ore shipments, mine exports, signal manifest |
| Streaming JSON Lines            | .jsonl         | 307    | 995.9 MB               | <code>documents.jsonl</code> manifest, high-frequency FFA ticks, trade logs         |
| High-Density Parquet & Archives | .parquet / .gz | 41     | 281.8 MB               | PortWatch AIS congestion, historical vessel tracking, high-frequency tick archives  |
| Spreadsheet Workbooks           | .xlsx          | 49     | 18.5 MB                | Commodity balances, IMF PortWatch expansions, broker models                         |
| Indexed Knowledge Entities      | knowledge/*    | 10,134 | 3,250.0 MB             | Documents passed through OCR, entity linking, and knowledge graph                   |

## Forensic Mapping: What Is Currently Stored

### 1. The Signal Group Maritime Intelligence ( `reports/signal/` )

Forensically verified & updated September 19, 2026: 2,369 files, 318.1 MB across 6 specialized directories.

Signal Ocean and Signal Maritime represent a dedicated commercial intelligence stream harvested via `scripts/scrapers/fetch_signal_reports.py` and `scripts/scrapers/download_signal_images.py`. This is **not** an external bookmark collection; it is a **fully localized, offline-readable analytical intelligence repository with continuous weekly automation**:

| Subdirectory / Component            | File Type                                                            | Count | Total Size | Description & Intelligence Scope                                                                                                                                                                                                                                                         |
|-------------------------------------|----------------------------------------------------------------------|-------|------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">monitors/</a>           | <a href="#">.md</a>                                                  | 249   | 1.15 MB    | <b>Clean Weekly Market Monitors:</b> Structured markdown reports covering Dry Bulk (119), Tankers (99), and Commodity Radars (31). Includes latest Week 36 & 37 2026 reports, Capesize/Panamax ballaster trends, Arabian Gulf VLCC net supply, tonne-day growth, and Brazilian C3 rates. |
| <a href="#">newsroom/</a>           | <a href="#">.md</a>                                                  | 189   | 1.48 MB    | <b>Analytical Research &amp; Market Deep Dives:</b> Multi-page analyses of US Gulf to China WTI crude arbitrage, Capesize El Niño rainfall divide, Red Sea tanker rerouting, Shell pool commercial management, and EU ETS/FuelEU decarbonization economics.                              |
| <a href="#">images/</a>             | <a href="#">.png</a> / <a href="#">.avif</a> / <a href="#">.jpeg</a> | 1,426 | 200.8 MB   | <b>Visual Quantitative Charts:</b> 100% of these images are actively linked inside the Markdown reports. Captures proprietary Signal Ocean time-series plots, supply curves, and freight rate benchmarks.                                                                                |
| <a href="#">html/</a>               | <a href="#">.html</a>                                                | 500   | 100.2 MB   | <b>Offline Provenance DOM Snapshots:</b> Full-page raw Webflow DOM captures averaging ~207 KB. Zero redirect stubs. Preserves original layout, publication timestamps, and author attributions.                                                                                          |
| <a href="#">signal_manifest.csv</a> | <a href="#">.csv</a>                                                 | 1     | 0.14 MB    | <b>Master Audit Registry:</b> 500 rows tracking slugs, URLs, sections ( <a href="#">monitors</a> vs <a href="#">newsroom</a> ), local file paths, categories, character counts, and stub-filter flags (66 empty press mentions safely skipped).                                          |
| <a href="#">pdfs/</a>               | <a href="#">.pdf</a>                                                 | 4     | 16.1 MB    | <b>Regulatory Whitepapers:</b> Referenced source whitepapers, including the <i>Fourth IMO GHG Study 2020 Executive Summary</i> (10.2 MB), <i>Energy Efficiency Tech for Ships</i> (5.9 MB), and <i>EU ETS Maritime Directives</i> .                                                      |

2. Shipbroker Weekly Market Reports ( [reports/shipbrokers/](#) )

A multi-firm archive spanning 2018 through 2026 with **3,452 PDFs (2.74 GB)** cataloged across 18 leading global shipbroking houses:

| Shipbroking Firm            | Slug / Directory                                | Report Count | Total Size | Date Range  | Primary Focus Areas                                         |
|-----------------------------|-------------------------------------------------|--------------|------------|-------------|-------------------------------------------------------------|
| Simpson Spence Young        | <a href="#">ssy</a>                             | 517          | 90.3 MB    | 2021 – 2026 | Capesize/Panamax dry bulk, tanker spot, Atlantic sentiment  |
| Xclusiv Shipbrokers         | <a href="#">xclusiv</a>                         | 265          | 424.9 MB   | 2021 – 2026 | S&P transaction benchmarks, asset valuations, macro outlook |
| Fearnleys                   | <a href="#">fearnleys</a>                       | 256          | 177.9 MB   | 2021 – 2026 | Tanker/gas freight, North Sea, Suezmax/VLCC weekly pulse    |
| Golden Destiny              | <a href="#">golden_destiny</a>                  | 252          | 209.9 MB   | 2021 – 2026 | Weekly S&P lists, newbuilding contracting, demolition sales |
| Intermodal Shipbrokers      | <a href="#">intermodal</a>                      | 250          | 324.8 MB   | 2021 – 2026 | Time charter assessment tables, bunker price tracking, S&P  |
| Affinity (Shipping)         | <a href="#">affinity</a>                        | 249          | 47.8 MB    | 2021 – 2026 | Tanker commercial reports, LNG chartering, macro shipping   |
| Advanced Shipping & Trading | <a href="#">advanced_shipping</a>               | 248          | 255.9 MB   | 2021 – 2026 | Secondhand vessel transactions, demolition prices, fixtures |
| Banchero Costa (bancosta)   | <a href="#">banchero_costa</a>                  | 243          | 248.9 MB   | 2021 – 2026 | Commodity trade flows, fleet age profiles, dry bulk supply  |
| Agora Shipbroking           | <a href="#">agora</a>                           | 212          | 174.6 MB   | 2021 – 2026 | Dry bulk freight indices, regional grain/coal movements     |
| Allied Shipbroking          | <a href="#">allied</a>                          | 204          | 357.4 MB   | 2021 – 2026 | Comprehensive dry bulk/tanker supply-demand, S&P benchmarks |
| Star Asia Shipbroking       | <a href="#">star_asia</a>                       | 191          | 163.3 MB   | 2021 – 2026 | Indian subcontinent demolition cash buyers, recycling rates |
| Carriers Chartering         | <a href="#">carriers</a>                        | 128          | 42.1 MB    | 2021 – 2026 | Handysize / Supramax spot fixtures, Black Sea / Med routes  |
| ISM Navigation              | <a href="#">ism</a>                             | 112          | 17.9 MB    | 2021 – 2026 | S&P market summaries, coastal trading, fleet updates        |
| Gibson Shipbrokers          | <a href="#">gibson</a>                          | 109          | 83.0 MB    | 2021 – 2026 | Crude/dirty tanker weekly outlooks, OPEC+ quotas, ton-miles |
| Lion Shipbrokers            | <a href="#">lion</a>                            | 44           | 78.6 MB    | 2021 – 2026 | Detailed vessel sale transactions with buyers/sellers named |
| Anchor Shipbroking          | <a href="#">anchor</a>                          | 30           | 3.5 MB     | 2022 – 2026 | Specialized Hellenic boutique fixtures & demo evaluations   |
| Clarksons Hellas            | <a href="#">clarksons</a>                       | 9            | 1.8 MB     | 2026        | Specialized Hellenic weekly circulars and orderbook reviews |
| Other / Regional Brokers    | <a href="#">other</a> / <a href="#">general</a> | 133          | 42.4 MB    | 2021 – 2026 | Regional brokers, commodity houses, boutique assessments    |
| SUBTOTAL                    | —                                               | 3,452        | 2,744.8 MB | 2021 – 2026 | Institutional Broker Intelligence Baseline                  |

NOTE

> In addition to the raw PDFs, Fearnleys desk intelligence is indexed into `data/derived/fearnleys\_broker\_comments.csv`, containing **\*\*11,731 individual desk comments\*\*** dating back to September 2018 across Tankers, Dry Bulk, Gas, and S&P. Furthermore, **\*\*176 structured Markdown circulars (1.6 MB)\*\*** are preserved in `reports/fearnleys/` (partitioned by year: `2024/`, `2025/`, `2026/`). In parallel, 138 multi-firm broker reports are normalized in `reports/broker\_reports/\*\*\*.md` across 14 top houses.

3. Hellenic Shipping News Spot & Sector Streams ( [reports/hellenic/](#) )

The primary Hellenic category ingest aggregates **14,122 total files (3.20 GB)**: - **Native Document PDFs: 3,965 PDFs (2.20 GB)** across three specialized sectors: - **Iron Ore Spot Market ( [reports/hellenic/iron\\_ore/](#) ) — 2,240 PDFs (1,351.4 MB)**: MMI (Metals Market Index) Daily Iron Ore Reports covering 62% Fe Qingdao, 65% Fe Carajas, lump/pellet premiums, and 35 Chinese port stocks. - **Ship Demolition & Recycling ( [reports/hellenic/demolition/](#) ) — 1,050 PDFs (652.9 MB)**: Scrap cash valuations from GMS, Best Oasis, and Star Asia across Chattogram, Alang, Gaddani, and Aliaga. - **Shipbuilding & Contracting ( [reports/hellenic/shipbuilding/](#) ) — 675 PDFs (197.2 MB)**: Yard orderbook logs, slot availability, and contracting benchmarks. - **Companion Multimodal Intelligence: 10,157 companion files (1.00 GB)** omitted by standard PDF-only scans: - **6,585 JPG Market Charts**: High-resolution price charts and smelter margin graphics. - **3,201 Full HTML Articles**: Complete editorial articles and fixture commentary. - **326 PNG Flow Graphics**: Route maps and capacity distribution diagrams.

4. Baltic Exchange Historical Intelligence Archive ( [reports/baltic/](#) )

- **Total Reports: 3,038 HTML files (14.0 MB)**
- **Coverage:** Complete historical archive of official Baltic Exchange fixture circulars, sector commentary articles, and route assessment reports across Capesize, Panamax, Supramax, Handysize, VLCC, and Clean Tanker segments.
- **Integration:** Directly parsed into [knowledge/trees/](#) and linked with the live index engine in [scripts/baltic\\_new\\_indices.py](#).

## 5. Drewry Maritime Intelligence ( [scripts/drewry\\_ais\\_pdfs/](#) & [reports/drewry/](#) )

- **AIS Vessel Class PDFs** ( [scripts/drewry\\_ais\\_pdfs/](#) ): **276 PDFs (501.9 MB)** of dense AIS tracking heatmaps and fleet positioning data (Aframax, Suezmax, VLCC, Capesize, Panamax, Supramax, MR, LR1, LR2).
- **Structured Opinion Briefs** ( [reports/drewry/](#) ): **547 Markdown reports (2.0 MB)** containing Drewry's analytical viewpoints, container market opinions, and dry bulk trade balance forecasts.
- **Rate Indices:** Full historical series of the Drewry World Container Index (WCI) in [data/indices/drewry\\_wci.csv](#).

## 6. Poten & Partners Tanker Opinions ( [reports/poten/](#) )

- **Total PDFs: 1,085 PDFs** (277.4 MB)
- **Extracted Markdown Opinions: 1,094 .md files**
- **Span: January 2005 through September 2026 (21-year unbroken time series)**
- **Coverage:** Weekly deep-dive essays examining crude tanker supply, clean product flows, sanction effects, Panama/Suez Canal diversions, dark fleet mechanics, and refinery restructuring.

## 7. Breakwave Advisors Research Engine ( [reports/drybulk/](#) , [reports/tankers/](#) , [reports/breakwave/](#) )

- **Dry Bulk Reports** ( [reports/drybulk/](#) ): **210 PDFs** (64.0 MB) spanning 2018–2026.
- **Tanker Reports** ( [reports/tankers/](#) ): **79 PDFs** (28.7 MB) spanning 2023–2026.
- **Market Insights & Special Presentations** ( [reports/breakwave/](#) ): **81 PDFs** (20.1 MB) + **3,190 HTML/Markdown insights**.
- **Visual Chart Repository: 15,150 high-resolution visual chart files (2.41 GB)** (10,064 PNGs, 4,602 JPGs, 365 JPEGs) capturing freight futures (FFA) forward curves, spot/futures spreads, and tanker/dry bulk indices.

## 8. Seabrokers Offshore Market Intelligence ( [data/reports/seabrokers/](#) & [reports/seabrokers/](#) )

- **Total PDFs: 97 PDFs** (570.3 MB)
- **Extracted Markdown Reports** ( [reports/seabrokers/](#) ): **97 .md circulars** partitioned by year ( [2020/](#) through [2026/](#) ).
- **Average Size:** 6.02 MB per report (high-fidelity monthly technical publications)
- **Coverage:** Platform Supply Vessels (PSV), Anchor Handling Tug Supply (AHTS), subsea construction vessels, and offshore drilling rig dayrates across the North Sea, Gulf of Mexico, West Africa, and Brazil.

## 9. Pilbara Ports Authority (PPA) Throughput Reports ( [scratch/ppa\\_pdf/](#) & [scratch/](#) )

- **Total PDFs: 492 PDFs** (50.6 MB) across [scratch/ppa\\_pdf/](#) (326 PDFs) and root [scratch/](#) (166 PDFs).
- **Coverage:** Monthly official port authority tonnage statistics from **Port Hedland** (BHP, FMG, Roy Hill) and **Port of Dampier** (Rio Tinto), tracking iron ore export volumes to China, Japan, and South Korea, berth calls, and vessel congestion.

## 10. Quantitative, Commodity & Geospatial Datasets ( [data/](#) )

The repository contains critical structured time-series datasets that power the analytical and mapping workstations:

1. [data/congestion/](#) — **19 files (214.4 MB)**: Daily IMF PortWatch port congestion series ( [portwatch\\_port\\_congestion.csv](#) ), global maritime disruption feeds, and compressed AIS historical congestion archives ( [.parquet](#) , [.gz](#) ).
2. [data/bunkers/](#) — **12 files (186.5 MB)**: Marine fuel benchmark price series (VLSFO, MGO, IFO380, LNG) across major bunkering hubs: Singapore, Rotterdam, Fujairah, and Houston.
3. [data/commodities/](#) — **1,231 files (68.8 MB)**: Primary tabular trade feeds: Brazil ComexStat exports, Pilbara Ports monthly throughput, China customs values/tonnages, US EIA energy shipments, and Newcastle coal loadings.
4. [data/futures/](#) & [data/equities/](#) — **29 files (67.3 MB)**: 16 FFA freight futures settlement curves (41.8 MB) and 13 shipping public equities price/volume time-series (25.5 MB).
5. [data/geospatial/](#) — **34+ files (64.2 MB)**: 12,060 ports gazetteer, live fleet AIS coordinates, 57,000 commercial vessel register, [voyage\\_history\\_packed.json](#) (4,537 tracked vessels, 357,000 voyage legs), [data/geospatial/history/](#) (permanent append-only monthly Parquet archives compressed with Zstandard zstd-7), and [data/views/signal/ais\\_history\\_active.json](#) (trajectory reconstruction engine enabling browser map playback).
6. [data/clarksons/](#) — **65 files (2.7 MB)**: 5 PDFs plus 48 JSONs, 8 CSVs, and 3 HTMLs including the master [gibson\\_all\\_reports\\_catalog.json](#) and market rate tables.
7. [docs/alibra\\_data/](#) — **70 files (1.8 MB)**: Tanker/dry bulk forward curve datasets, time charter poller logs, and validation matrices.

## 11. Macro Financial Regulatory Statements & Institutional Research

- **CFTC Commitments of Traders** ( [data/cftc\\_statements/raw\\_pdf/](#) ): **138 PDFs** (37.1 MB) covering managed money, producer/merchant, and swap dealer net long/short positions in freight, crude oil, and commodity derivatives.
- **Amplify ETF Regulatory Documents** ( [docs/](#) ): **7 PDFs** (11.1 MB) including BDRY & BWET Prospectuses, FactSheets, and SEC Form 10-Q statements.
- **UN Comtrade Specifications** ( [docs/research/](#) ): **2 PDFs** (0.4 MB) data schemas for bilateral customs trade reconciliation.

## 12. Foundational Maritime Economics Textbooks ( [reports/\\*.pdf](#) )

A curated library of **12 foundational reference textbooks (118.3 MB)** directly ingested into [scripts/process\\_knowledge.py](#) to establish the semantic baseline for the knowledge graph:

1. **Maritime Economics, 3rd Edition** — Martin Stopford (*The definitive 800-page treatise on shipping market cycles*)
2. **Maritime Economics: A Macroeconomic Approach** — Elias Karakitsos & Lambros Varnavides
3. **The International Handbook of Shipping Finance: Theory and Practice** — Manolis G. Kavussanos & Ilias D. Visvikis
4. **Lloyd's Maritime Atlas of World Ports and Shipping Places, 24th Edition**
5. **The Business of Shipping** — Lane C. Kendall

6. The World's Key Industry: History and Economics of International Shipping — G. Harlaftis, S. Tenold, J. Valdaliso

7. The Sea and Civilization: A Maritime History of the World — Lincoln Paine

8. Shipping Business Unwrapped — Okan Duru

9. The Shipping Man — Matthew McCleery

10. Quantitative Modelling of Shipping Freight Rates: Developments in the Past 20 Years (2022)

11. Predictability of Second-Hand Bulk Carriers with a Novel Hybrid Model

12. Types of Ships and Maritime Cargo Handling

## Processing & Weekly Ingestion Pipeline

### Automated Inflow Architecture (23 GitHub Actions Workflows)

The repository operates **23 orchestrated GitHub Actions workflows**. The active document and data ingestion engines run on automated schedules:

```
flowchart TD
    subgraph MultiBroker [Shipbroker & Commercial Intelligence]
        W1["broker_reports_weekly.yml\nMon/Wed/Fri 07:00 UTC\nFri 18:00 UTC"]
        W2["broker_voice_sync.yml\nDaily 07:20 & 17:20 UTC"]
        W3["fearnleys_weekly.yml\nWed 15:45 & Thu 07:45 UTC"]
        W_SIG["signal_reports_weekly.yml\nFri 18:30 & Mon 08:30 UTC"]
    end

    subgraph SpecializedSectors [Commodity & Sector Engines]
        W4["poten_drewry_weekly.yml\nFri 17:00 UTC"]
        W5["report_ingest.yml\nMon-Fri 08:00, 12:00, 16:00 UTC\nExtended: 09:30 UTC"]
        W6["offshore_seabrokers_monthly.yml\nMon/Thu 06:30 UTC + 1st-8th daily"]
        W7["upstream_commodity_flows.yml\nMon 06:00 UTC (24 scrapers)"]
        W8["usda_weekly.yml\nThu 15:00 UTC"]
    end

    subgraph MarketFeeds [Real-Time Indices & Macro]
        W9["alibra_poller.yml\nDaily 07:00 & 16:00 UTC"]
        W10["baltic_new_indices_update.yml\nMon-Fri 4x daily"]
        W11["etf_holdings_update.yml\nMon-Fri 14:00 UTC"]
    end

    subgraph KnowledgeEngine [RAG & Semantic Indexing]
        K1["daily_knowledge_update.yml\nDaily 15:30 UTC"]
        K2["daily_brief.yml\nMon-Fri 12:00 UTC"]
    end

    W1 --> D1["reports/shipbrokers/ & reports/broker_reports/<year>/<broker>/"]
    W2 --> D2["data/derived/fearnleys_broker_comments.csv"]
    W3 --> D3["data/derived/time_charter_rates_fearnleys.csv & reports/fearnleys/<year>/"]
    W_SIG --> D_SIG["reports/signal/ (monitors, newsroom, images)"]
    W4 --> D4["reports/poten/, reports/drewry/, scripts/drewry_ais_pdfs/"]
    W5 --> D5["reports/hellenic/, reports/drybulk/, reports/breakwave/"]
    W6 --> D6["data/reports/seabrokers/ & reports/seabrokers/<year>/"]
    W7 --> D7["data/commodities/ (PPA, Miners, BPS, ComexStat)"]
    W8 --> D8["data/commodities/ (USDA FGIS Inspections & Queues)"]

    D1 --> K1
    D4 --> K1
    D5 --> K1
    D_SIG --> K1
    K1 --> G1["knowledge/trees/ & documents.json\n(10,134 indexed entities)"]
    G1 --> K2
```

## Forensic Stream-by-Stream Verification

| Stream                        | Update Cadence    | Active Workflow                                 | Trigger Schedule                                      | Scraper Scripts Executed                                                                                                                                                                     | Downstream Knowledge / RAG Destination                                                                                                                                         |
|-------------------------------|-------------------|-------------------------------------------------|-------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. Multi-Broker Reports       | Weekly            | <a href="#">broker_reports_weekly.yml</a>       | Mon/Wed/Fri 07:00 UTC + Fri 18:00 UTC (4x/wk)         | <a href="#">fetch_hsn_shipbrokers.py</a> , <a href="#">update_intermodal_tc_rates.py</a> , <a href="#">fetch_gibson_weekly.py</a> , <a href="#">extract_demolition_pdfs.py</a>               | Parsed to <a href="#">reports/broker_reports/&lt;year&gt;/&lt;broker&gt;/*.md</a> ; daily indexed by <a href="#">process_knowledge.py</a>                                      |
| 2. Broker Voice & Catalog     | Daily (2x/day)    | <a href="#">broker_voice_sync.yml</a>           | Daily 07:20 & 17:20 UTC                               | <a href="#">daily_fearnleys_sync.py</a> , <a href="#">build_comment_chunks.py</a> , <a href="#">fetch_gibson_catalog.py</a> , <a href="#">check_broker_voice_fresh.py</a>                    | Syncs Gibson catalog + Fearnleys desk comments (11.7k comments)                                                                                                                |
| 3. Fearnleys Fixtures & TC    | Weekly            | <a href="#">fearnleys_weekly.yml</a>            | Wed 15:45 UTC & Thu 07:45 UTC                         | <a href="#">fetch_fearnleys_tc.py</a> , <a href="#">fetch_fearnleys_reports.py</a> , <a href="#">daily_fearnleys_sync.py</a> , <a href="#">build_fearnleys_cache.py</a>                      | Updates <a href="#">time_charter_rates_fearnleys.csv</a> and <a href="#">reports/fearnleys/&lt;year&gt;/*.md</a>                                                               |
| 4. The Signal Group           | Weekly (2x/wk)    | <a href="#">signal_reports_weekly.yml</a>       | Fri 18:30 UTC & Mon 08:30 UTC                         | <a href="#">fetch_signal_reports.py</a> , <a href="#">download_signal_images.py</a>                                                                                                          | Ingests to <a href="#">reports/signal/monitors/*.md</a> (249) and <a href="#">reports/signal/newsroom/*.md</a> (189) with 1,426 local chart images                             |
| 5. Hellenic Spot & Sector     | Daily / Weekly    | <a href="#">report_ingest.yml</a>               | Mon–Fri 09:30 UTC ( extended )                        | <a href="#">hellenic_scraper.py --category all</a> (Iron Ore MMi, Demolition, Shipbuilding, Dry/Wet Charter)                                                                                 | Stored in <a href="#">reports/hellenic/</a> ; daily OCR/indexed by <a href="#">process_knowledge.py</a>                                                                        |
| 6. Poten Tanker Opinions      | Weekly            | <a href="#">poten_drewry_weekly.yml</a>         | Every Friday 17:00 UTC                                | <a href="#">fetch_poten_direct.py</a>                                                                                                                                                        | Extracted to <a href="#">reports/poten/*.md</a> ; indexed in <a href="#">knowledge/manifests/documents.jsonl</a>                                                               |
| 7. Drewry AIS & WCI           | Weekly            | <a href="#">poten_drewry_weekly.yml</a>         | Every Friday 17:00 UTC                                | <a href="#">fetch_drewry_wci.py</a> , <a href="#">fetch_drewry_ais_weekly.py</a> , <a href="#">fetch_drewry_opinions_incremental.py</a>                                                      | PDFs to <a href="#">scripts/drewry_ais_pdfs/</a> , Markdown to <a href="#">reports/drewry/</a> , CSV to <a href="#">drewry_wci.csv</a>                                         |
| 8. Breakwave Research         | Bi-Weekly / Daily | <a href="#">report_ingest.yml</a>               | Mon–Fri 08:00, 12:00, 16:00 UTC ( core ) + 09:30 UTC  | <a href="#">breakwave_scraper.py</a> , <a href="#">breakwave_insights_scraper.py</a>                                                                                                         | Outputs to <a href="#">reports/drybulk/</a> , <a href="#">reports/tankers/reports/breakwave/</a> ; daily indexed                                                               |
| 9. Seabrokers Offshore        | Monthly           | <a href="#">offshore_seabrokers_monthly.yml</a> | Mon & Thu 06:30 UTC + 1st–8th monthly daily 08:00 UTC | <a href="#">fetch_seabrokers_reports.py --auto</a> , <a href="#">build_offshore_cache.py</a>                                                                                                 | PDFs to <a href="#">data/reports/seabrokers/</a> & markdown to <a href="#">reports/seabrokers/&lt;year&gt;/*.md</a> , parsed into <a href="#">offshore_market_summary.json</a> |
| 10. Pilbara Ports (PPA)       | Monthly           | <a href="#">upstream_commodity_flows.yml</a>    | Every Monday 06:00 UTC                                | <a href="#">fetch_australia_ppa.py</a>                                                                                                                                                       | Updates monthly throughput tables in <a href="#">data/commodities/pilbara_monthly_exports.csv</a>                                                                              |
| 11. USDA Grain Flows          | Weekly            | <a href="#">usda_weekly.yml</a>                 | Every Thursday 15:00 UTC                              | <a href="#">fetch_usda_grains.py</a> , <a href="#">fetch_usda_fas_exports.py</a> , <a href="#">fetch_usda_grain_queues.py</a> , <a href="#">scripts/acquire/backfill_fgis_inspections.py</a> | Updates <a href="#">usda_ytd_grain_inspections_top20.csv</a> and vessel loading queues                                                                                         |
| 12. Miner Filings (SEC/ASX)   | Quarterly         | <a href="#">upstream_commodity_flows.yml</a>    | Every Monday 06:00 UTC                                | <a href="#">fetch_major_miners_production.py</a> , <a href="#">verify_miners_provenance.py</a>                                                                                               | Updates <a href="#">major_miners_quarterly_production.csv</a> verified against EDGAR/ASX filings                                                                               |
| 13. Alibra TC Assessments     | Daily (2x/day)    | <a href="#">alibra_poller.yml</a>               | Daily 07:00 & 16:00 UTC                               | <a href="#">alibra_poller.py --integrate</a> , <a href="#">integrate_alibra_feed.py</a>                                                                                                      | Time charter assessments integrated into fleet freight models                                                                                                                  |
| 14. Baltic Exchange Indices   | Daily (4x/day)    | <a href="#">baltic_new_indices_update.yml</a>   | Mon–Fri 10:30, 14:00, 19:00, 22:00 UTC                | <a href="#">baltic_new_indices.py --repo</a> .                                                                                                                                               | Live BDI, BCI, BPI, BSI freight indices updated throughout the trading day                                                                                                     |
| 15. ETF Holdings (BDRY/BWET)  | Daily             | <a href="#">etf_holdings_update.yml</a>         | Mon–Fri 14:00 UTC                                     | <a href="#">update_etf_holdings.py</a> , <a href="#">build_provenance_manifest.py</a>                                                                                                        | Daily live contracts, NAV, and weightings from Amplify Firestore API                                                                                                           |
| 16. Daily Knowledge Pass      | Daily             | <a href="#">daily_knowledge_update.yml</a>      | Daily 15:30 UTC                                       | <a href="#">process_knowledge.py</a> , <a href="#">check_breakwave_freshness.py</a> , <a href="#">validate_knowledge.py</a>                                                                  | Ingests new reports, runs OCR + LLM entity linking, updates <a href="#">documents.jsonl</a>                                                                                    |
| 17. Institutional Daily Brief | Daily             | <a href="#">daily_brief.yml</a>                 | Mon–Fri 12:00 UTC                                     | <a href="#">generate_brief.py</a>                                                                                                                                                            | Compiles multi-broker sentiment, freight curves, and macro drivers                                                                                                             |

Semantic Knowledge Graph Structure ( [knowledge/manifests/documents.jsonl](#) )

The repository maintains an indexed semantic knowledge manifest containing **10,134 nodes**:

| Document Type / Source | Indexed Nodes | Primary Focus                                                                            |
|------------------------|---------------|------------------------------------------------------------------------------------------|
| hellenic               | 3,200         | Daily MMi Iron Ore indices, global demolition scrap pricing, shipbuilding contracting    |
| breakwave_insights     | 3,190         | Macro freight insights, dry bulk supply/demand commentary, commodity dynamic notes       |
| baltic                 | 2,213         | Baltic Exchange fixture notes, daily market commentary across Capesize, Panamax, Tankers |
| poten                  | 1,094         | 21-year historical archive of weekly tanker opinions (2005–2026)                         |
| breakwave              | 289           | Bi-weekly published Dry Bulk & Tanker reports                                            |
| broker_reports         | 136           | Deep-parsed weekly shipbroker circulars (SSY, Fearnleys, Allied, Intermodal)             |
| book                   | 12            | Foundational textbooks (Martin Stopford, Karakitsos, Kavussanos, Lloyd's Atlas, etc.)    |
| TOTAL                  | 10,134        | Fully linked semantic knowledge graph                                                    |

## Architecture & Storage Optimization Strategy

### 1. Git Repository Scalability ( .gitignore Guardrails)

To prevent repository bloat and respect GitHub's recommended repository size limits, heavy static binary PDF archives are intentionally isolated in `.gitignore` : - `reports/shipbrokers/**/*.pdf` (3,452 PDFs, 2.74 GB — *local/runner disk cache*) - `reports/poten/**/*.pdf` (1,085 PDFs, 277 MB — *local/runner disk cache*) - `scripts/drewry_ais_pdfs/` (276 PDFs, 501 MB — *local/runner disk cache*) - `scratch/ppa_pdf/` & `scratch/*.pdf` (492 PDFs, 50.6 MB — *local/runner disk cache*)

Meanwhile, all **extracted structured Markdown** ( `reports/broker_reports/<year>/<broker>/*.md` , `reports/fearnleys/<year>/*.md` , `reports/seabrokers/<year>/*.md` , `reports/signal/monitors/*.md` ), **JSON data indices** ( `data/derived/` ), and **clean text documents** ( `knowledge/docs/` ) are tracked in version control, ensuring 100% reproducibility of the user interface without carrying gigabytes of redundant static binary blobs in git history.

### 2. Canonical Directory Structure & Separation of Concerns

Following the repository architecture refactor: - **Scripts Centralization** ( `scripts/` ): All Python scripts reside exclusively in `scripts/` organized by domain ( `acquire/` , `gap_fill/` , `experiments/` , `cargo/` , `fearnleys/` , `geospatial/` , etc.). Zero test files and zero CSV data files reside in `scripts/` . - **Single-Tier Test Organization** ( `tests/` ): All test suites reside directly in `tests/test_*.py` with uniform execution depth ( `parent.parent == REPO_ROOT` ), with `sys.path` injection in `tests/conftest.py` . - **Hierarchical Broker Taxonomies**: Broker reports are sorted into `<year>/<broker>/*.md` across 14 global houses. Fearnleys and Seabrokers reports are partitioned cleanly by publication year. - **Reference & Documentation Archiving**: Unreferenced artifacts, prompt ledgers, and UI inspiration are archived into `docs/reference/` ( `megaprompts/` , `inspiration/` , `logs/` ), `docs/screenshots/` , and `docs/research/` . - **Knowledge Manifest Invariance**: Foundational academic textbooks are retained at `reports/*.pdf` matching immutable relative paths in `knowledge/manifests/manifest.jsonl` .

### 3. Permanent AIS Telemetry Archive ( `data/geospatial/history/` )

- Historical vessel telemetry is partitioned by month into Zstandard-compressed Parquet files ( `ais_positions_YYYY_MM.parquet` , level 7 compression).
- Seeded with 9,078 authentic observations across 8,857 hulls.
- Idempotently deduplicated on ( `imo` , `timestamp` ) with Null Island filtering.
- Reconstructed active voyage paths are compiled into `data/views/signal/ais_history_active.json` for interactive browser Leaflet map playback.

### 4. Knowledge Engine Indexing ( `knowledge/manifests/documents.jsonl` )

The processing engine converts incoming PDFs and web reports into a compact, searchable knowledge graph: - **10,134 indexed documents** with explicit source lineage. - **10,134 semantic syntax trees** ( `knowledge/trees/` ) recording entity hierarchies. - **Full-text searchability** without opening raw PDF streams. - **Fail-soft OCR bounds**: Maximum 16 pages per linked PDF, maximum 6 pages of heavy OCR per run, ensuring CI/CD runs never timeout or hang on dense scans.

## Summary of Findings

- Total Machine Footprint: 18,858 PDFs (11.32 GB)** across canonical directories and active agent worktrees; > **51,000 multimodal files (12.10 GB)** across the full repository.
- Canonical Core Collection: 9,905 PDFs (6.49 GB)** organized across 18 major shipbroking firms, 3 Hellenic commodity streams, 21 years of Poten tanker opinions, Drewry AIS tracking, Breakwave research, Pilbara throughput reports, and 12 foundational maritime economics textbooks.
- The Signal Group Intelligence Suite: 2,369 files (318.1 MB)** comprising 249 Weekly Market Monitors (.md), 189 Market Newsroom Deep Dives (.md), 1,426 local analytical chart graphics (.png), 500 raw HTML provenance snapshots, and 4 regulatory whitepapers.
- Multimodal Expansion: 3,038 Baltic Exchange HTML fixture reports, 547 Drewry Markdown opinion briefs, 176 Fearnleys Markdown circulars, and over 25,000 localized visual chart graphics** in `reports/hellenic/` and `reports/breakwave/` .
- Quantitative & AIS Datasets**: Dedicated time-series and AIS matrices in `data/` covering IMF PortWatch port congestion (214 MB), global marine bunker fuel benchmarks (186 MB), commodity balances (69 MB), FFA freight futures curves (42 MB), global fleet coordinates (63 MB), and append-only Parquet AIS telemetry archives ( `data/geospatial/history/` ).
- Automated Pipeline Health: 23 orchestrated workflows** running reliably across GitHub Actions, with 17 active data and document pipelines continuously updating rates, fixtures, desk comments, throughput statistics, Signal Ocean intelligence, and RAG entity trees.